

Generative Adversarial Networks (GANs)

Semester 2, 2022

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Demo

- <https://thispersondoesnotexist.com/>

Outline

- GAN architecture
- Evaluating GANs
- Conditional GANs

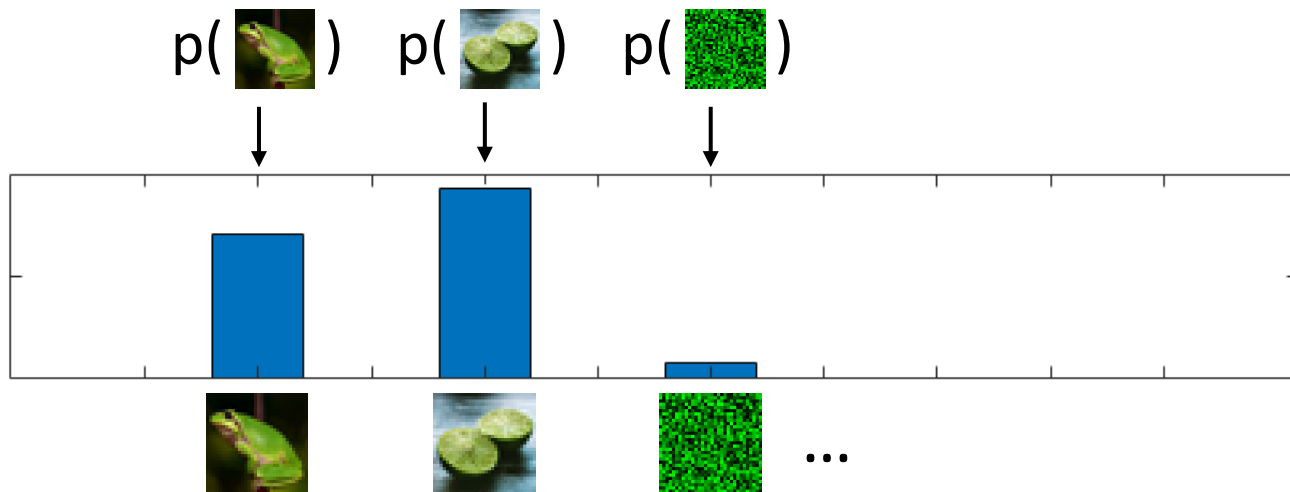
Learning Outcomes

- Explain the architecture and objective function of a standard GAN
- Explain common pitfalls in GANs, and how GANs can be evaluated
- Explain common architectures for conditional GANs

GAN architecture

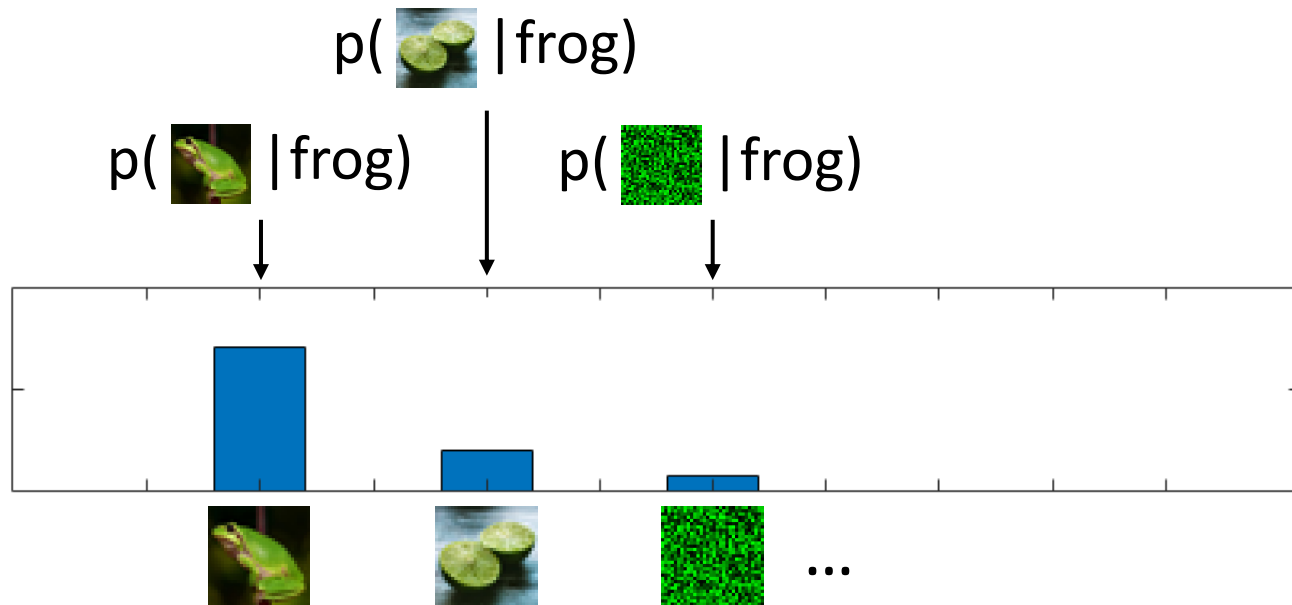
(Unconditional) generative model

- Output is a probability distribution $p(x)$
- What is the probability that this is an image?



(Conditional) generative model

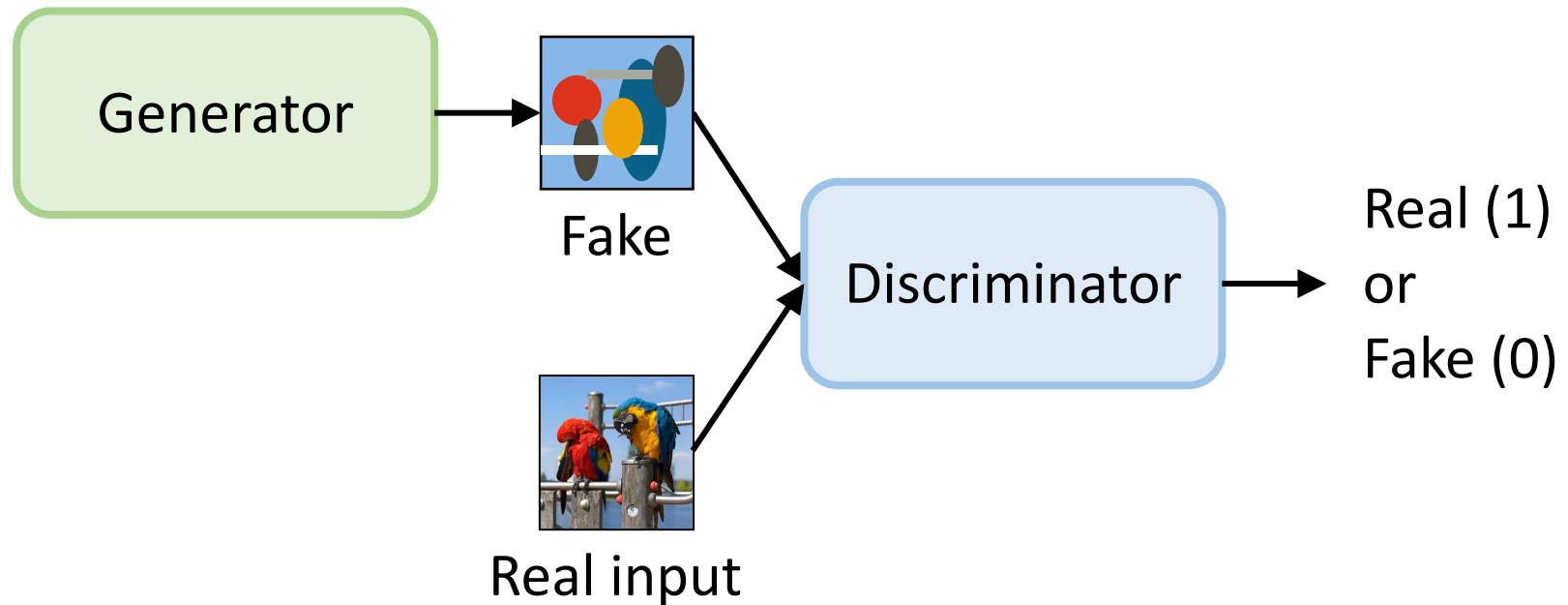
- Input is a label
- Output is a probability density function over images $p(x|y)$



GANs

- **Generative Adversarial Networks (GANs)** are neural networks that learn to generate instances from a particular distribution (e.g., images of faces)
- Actually consist of two neural networks: a **generator** and a **discriminator**
- Training involves a sort of competition between the two networks

GAN architecture

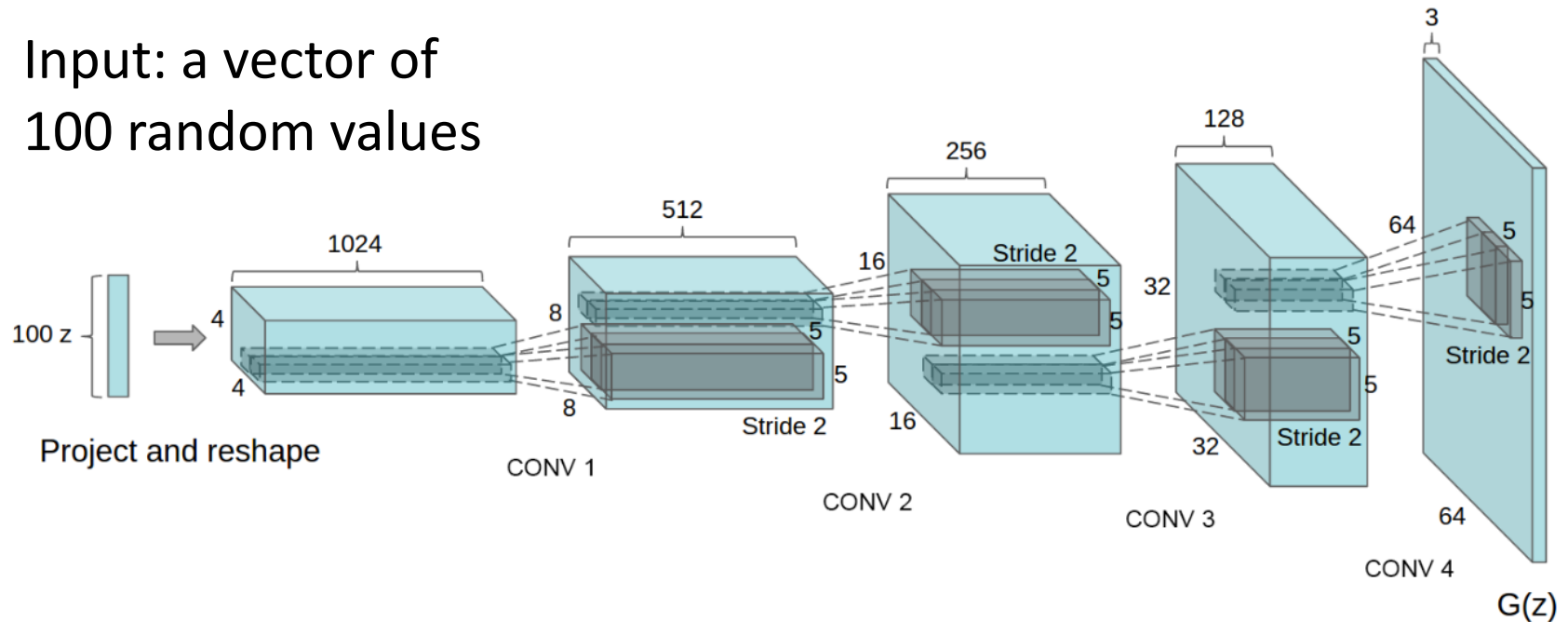


Generator

- GAN Generator doesn't actually learn the probability distribution $p(x)$, but learns to sample from it
- Generator input is a latent variable z with a simple prior (e.g., uniform random or standard normal)
- Generator output is an image
- Generator network learns a function to map $p(z)$ to a distribution that approximates $p(x)$

Generator architecture example

Input: a vector of
100 random values



Output: 64 x 64
pixel colour image

Discriminator

- Discriminator learns to identify real inputs vs. fake inputs created by generator
- Neural network classifier with two output classes (real, fake)
- Architecture depends on task: e.g., for images the discriminator might be a CNN with several convolutional layers, followed by softmax or Sigmoid

Training

- The networks are trained together on a combination of real data $\mathbf{x}$ and generator input $\mathbf{z}$
- Given a generator G and discriminator D :
 - Discriminator's goal is to correctly classify real vs. fake
 - Discriminator wants to maximize $D(\mathbf{x})$ and minimize $D(G(\mathbf{z}))$
 - Generator's goal is to fool the Discriminator
 - Generator wants to maximize $D(G(\mathbf{z}))$
- Can treat this as a zero-sum game with the goal of finding equilibrium between G and D

Training objective

- GAN training objective is a minimax game:

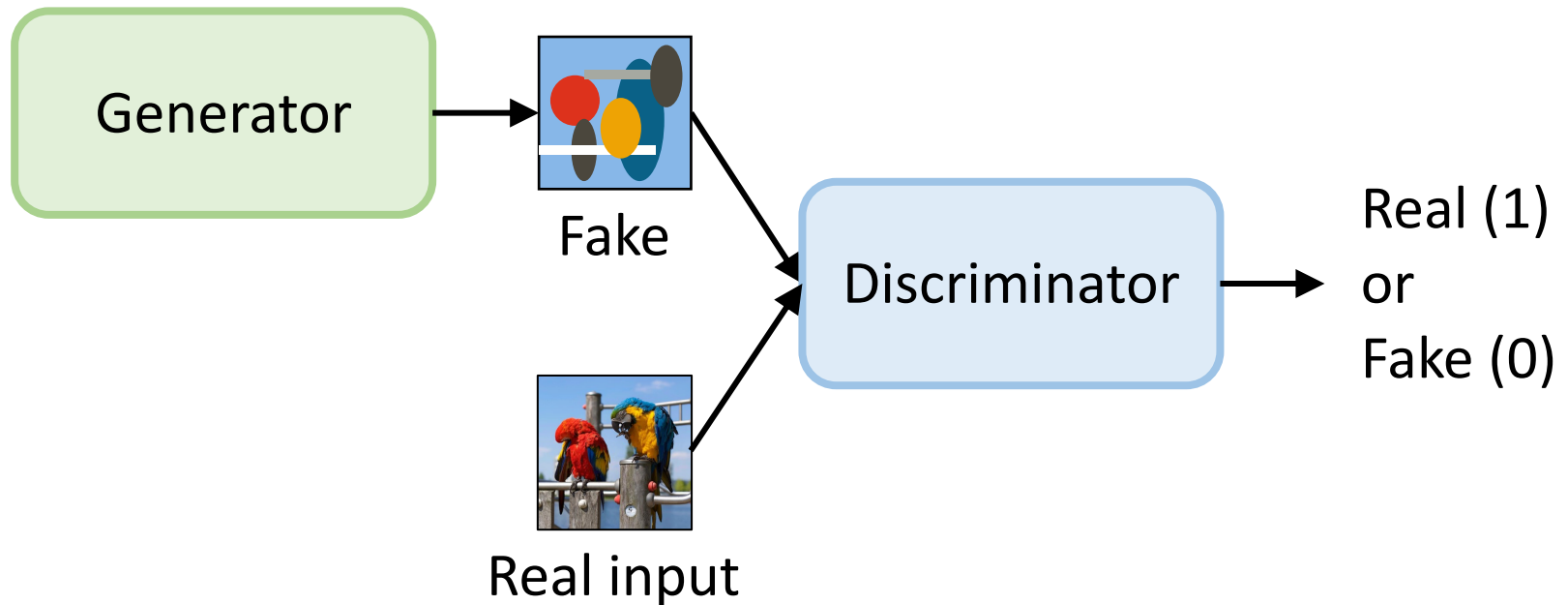
$$\min_G \max_D (E_{x \sim p_{data}} [\log(D(x))] + E_{z \sim p(z)} [\log(1 - D(G(z)))])$$

Discriminator's response to real images	1 - Discriminator's response to fake images produced by Generator
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The Discriminator tries to maximize this by learning weights for D that will give $D(x)=1$ for real images and $D(G(z))=0$ for fake images

The Generator tries to minimize this by learning weights that will give $D(G(z)) = 1$

Training



Training

- If the discriminator is too good:
 - Easily rejects all fake inputs
 - Not much information to train the generator
- If the discriminator is too bad:
 - Easily confused by fake inputs that don't look real
 - Generator will learn a poor solution
- Training can be difficult – hard to find a balance between discriminator and generator

Demo

- <https://poloclub.github.io/ganlab/>

Summary

- GAN is a pair of networks trained together: generator creates images based on latent input z , discriminator judges whether images are real vs. fake
- Objective function is a competition in which generator tries to fool discriminator
- Generator doesn't learn $p(x)$ (distribution of real images) but does learn to sample this distribution

Evaluating GANs

GAN evaluation

- GAN equilibrium does not necessarily mean the GAN has found a good solution
- How to tell if a GAN has learned? Ideally:
 - Outputs should not be identical to inputs (memorised training data)
 - Outputs should look like inputs (look “real” and not “fake”)
 - Outputs should be as diverse as real data (avoid **mode collapse** = the generator only creates one or a few outputs)

PROGRESSIVE GROWING OF GANs FOR IMPROVED QUALITY, STABILITY, AND VARIATION

Tero Karras
NVIDIA

Timo Aila
NVIDIA

Samuli Laine
NVIDIA

Jaakko Lehtinen
NVIDIA
Aalto University



<https://www.youtube.com/watch?v=G06dEcZ-QTg>

Memorisation?

GAN
output:

3 nearest
neighbours
in training
dataset



Realism?



Realism?



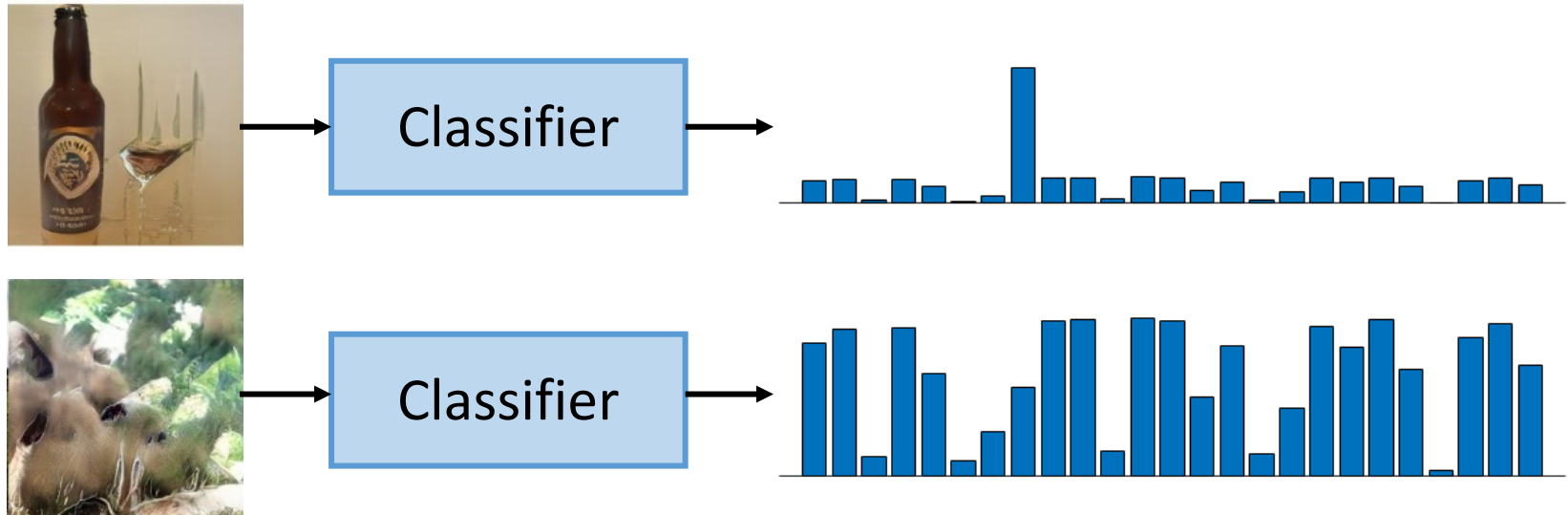
Realism?



Evaluating realism

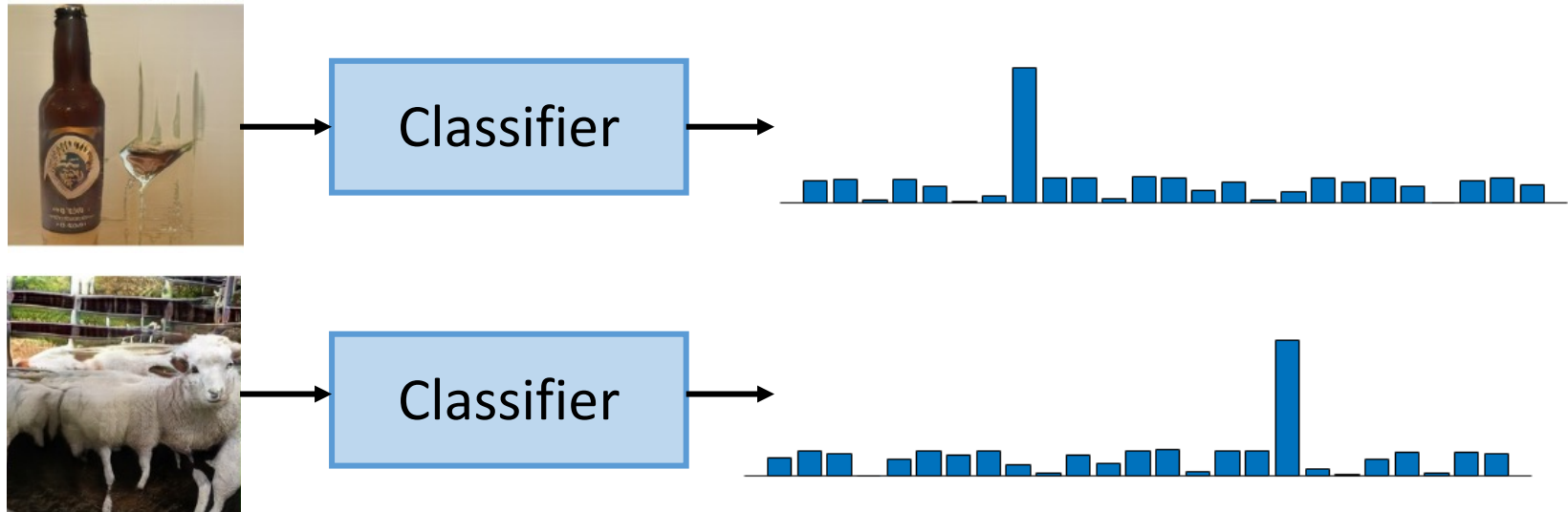
- How to evaluate realism?
- Gold standard: human evaluation (but this is slow and expensive)
- Automatic methods compare responses of an image classifier (e.g., a CNN trained on ImageNet) to real vs. GAN-generated images

Inception score



- Within a class, all images should be confidently classified with the correct label

Inception score



- Within a class, all images should be confidently classified with the correct label
- Across classes, the GAN should produce a wide variety of confidently-classified images

Inception score

- Advantages
 - Automatic, efficient
 - Neural network responses correlate with human judgements of image quality
- Disadvantages
 - Doesn't require high diversity within categories
 - Sensitive to noise, adversarial images

Diversity?

- The GAN isn't just memorizing training examples
- But does it capture *all* of the diversity in the training set?
 - How would you measure this?

Birthday paradox

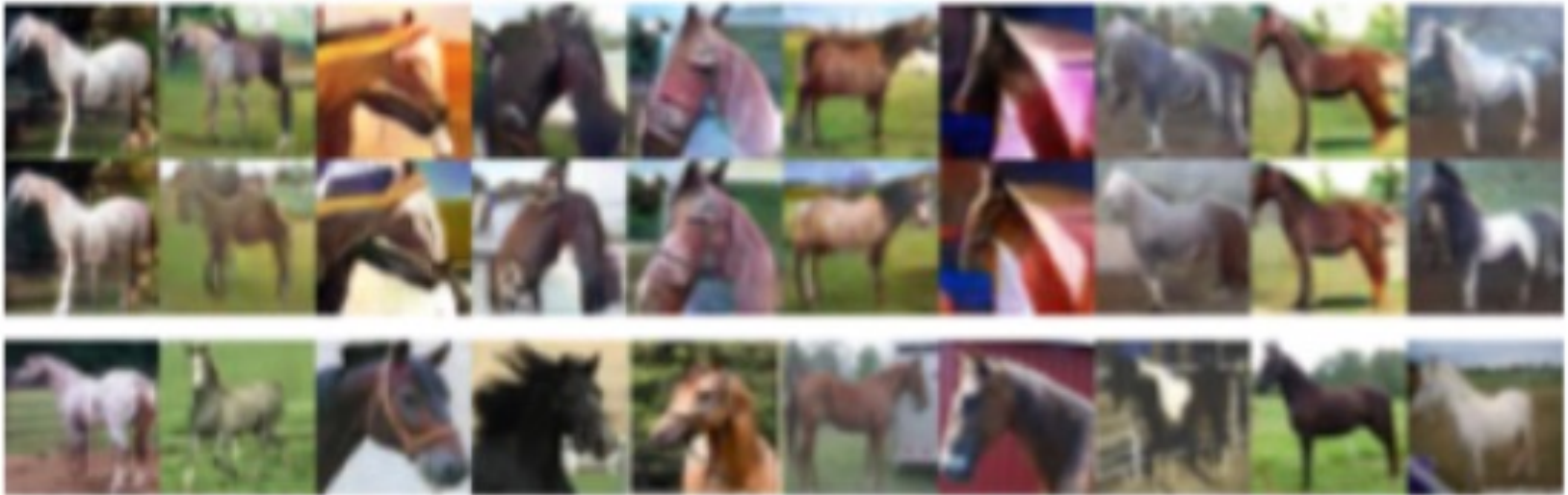
- What are the odds that someone else in this subject has the same birthday you do?
 - About 59% ($= 1 - (364/365)^{324}$)
- What are the odds that any two people in this subject share a birthday?
 - Close to 100%
- What's the smallest class size that has at least a 50/50 chance of two people sharing a birthday?
 - 23

Birthday paradox for GANs

- Arora, Risteski, & Zhang (2018)
- Suppose a generator that can produce N discrete outputs, all equally likely
- Experiment: take a small sample of s outputs and count duplicates
 - The odds of observing duplicates in a sample of size s can be used to compute N
 - A sample of about $\sqrt{N}$ outputs is likely to contain at least one pair of duplicates

Duplicates and diversity

- Example duplicates (and 1-NN in training dataset):



- Most GANs tested produced about the same diversity (number of different images) as was in their training set

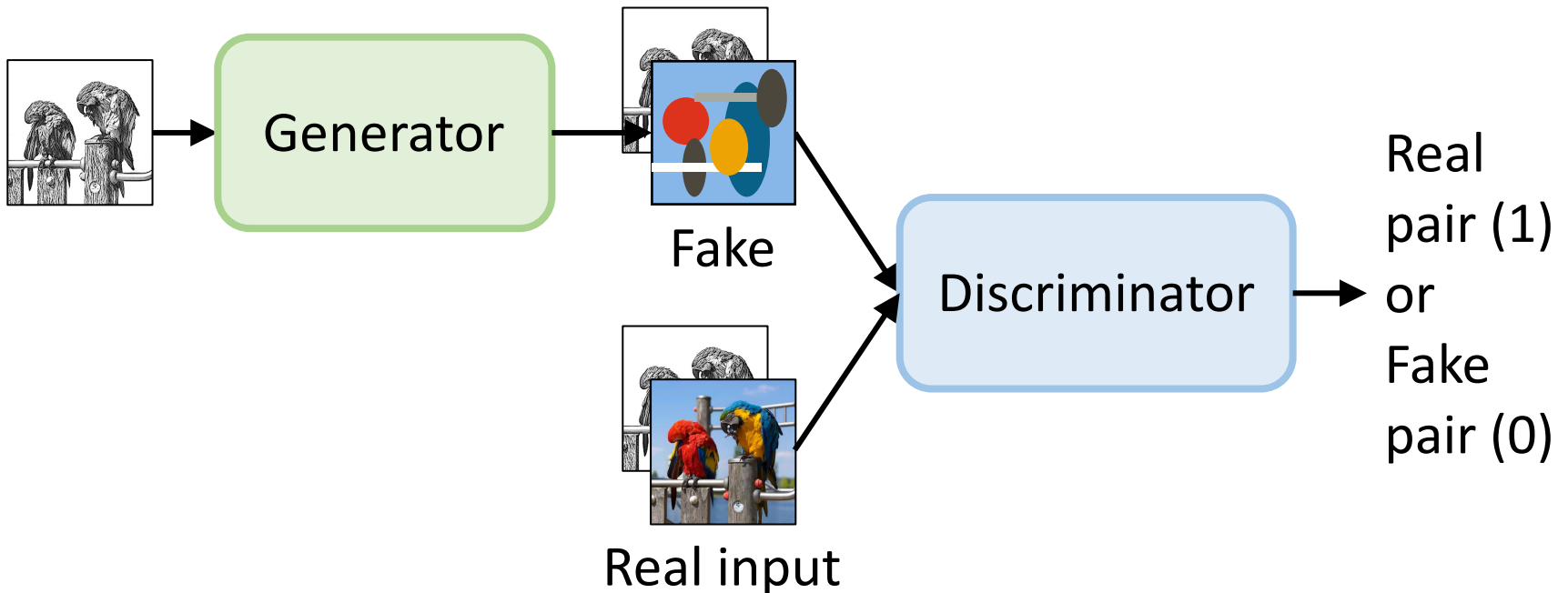
Summary

- GAN evaluation is important, because successful training does not necessarily mean the generator's output is similar to $p(x)$
- Generally check for:
 - Memorisation
 - Realism
 - Diversity

Conditional GANs

Conditional GANs

- Conditional model: learn $p(x|y)$ rather than $p(x)$
- Both Discriminator and Generator take y as additional input

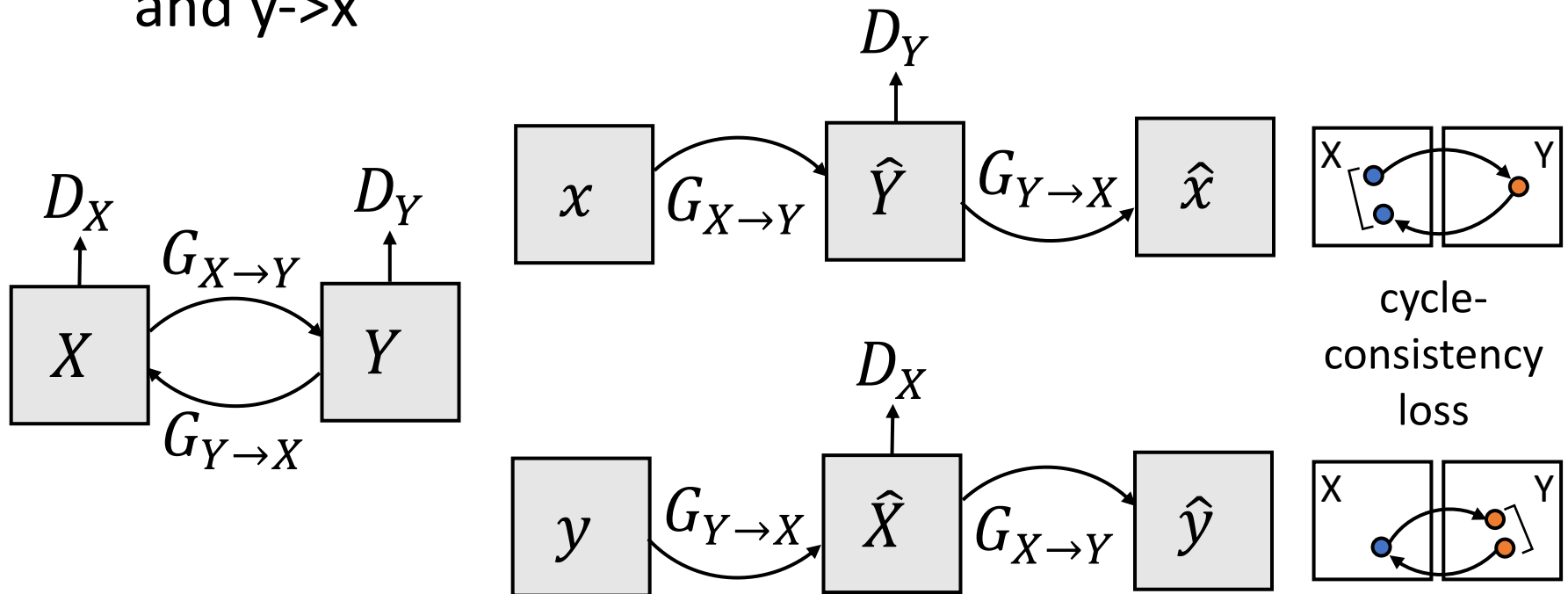


Demo

- <https://affinelayer.com/pixsrv/>

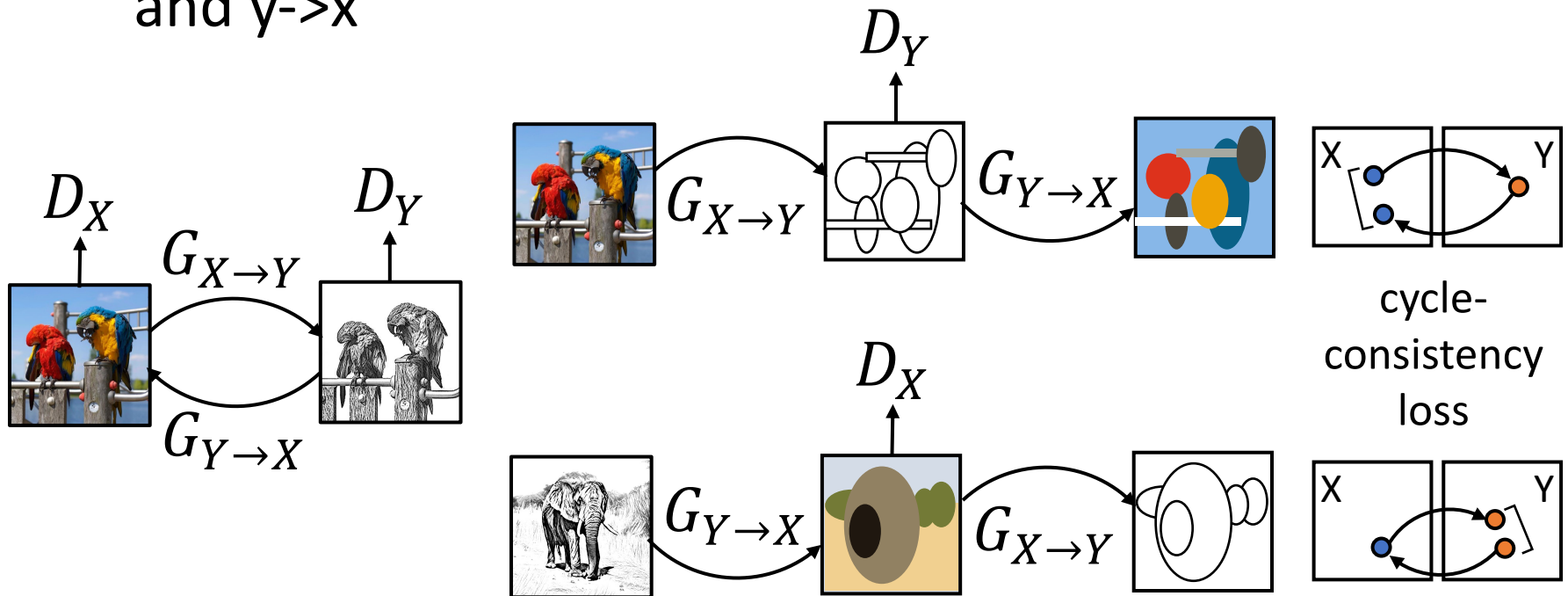
Conditional GANs

- What if you don't have a dataset of real x-y pairs?
- CycleGAN: train a pair of Generators to map $x \rightarrow y$ and $y \rightarrow x$



Conditional GANs

- What if you don't have a dataset of real x-y pairs?
- CycleGAN: train a pair of Generators to map $x \rightarrow y$ and $y \rightarrow x$



Example: CycleGAN



apple $\rightarrow$ orange



orange $\rightarrow$ apple

Image: <https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix>

Summary

- Conditional GAN learns to sample $p(x|y)$ (images conditional on y)
- If x - y pairs are available, can use the standard GAN architecture with additional input y
- If x - y pairs are not available, one option is CycleGAN – learns to transform samples from $x \rightarrow y$ and from $y \rightarrow x$

Summary

- Advantages
 - GANs can generate samples from complex probability distributions without actually representing the distribution
- Disadvantages
 - Can be unstable / hard to train
 - Difficult to evaluate
 - Even models that don't show complete mode collapse tend to have lower-than-desired diversity